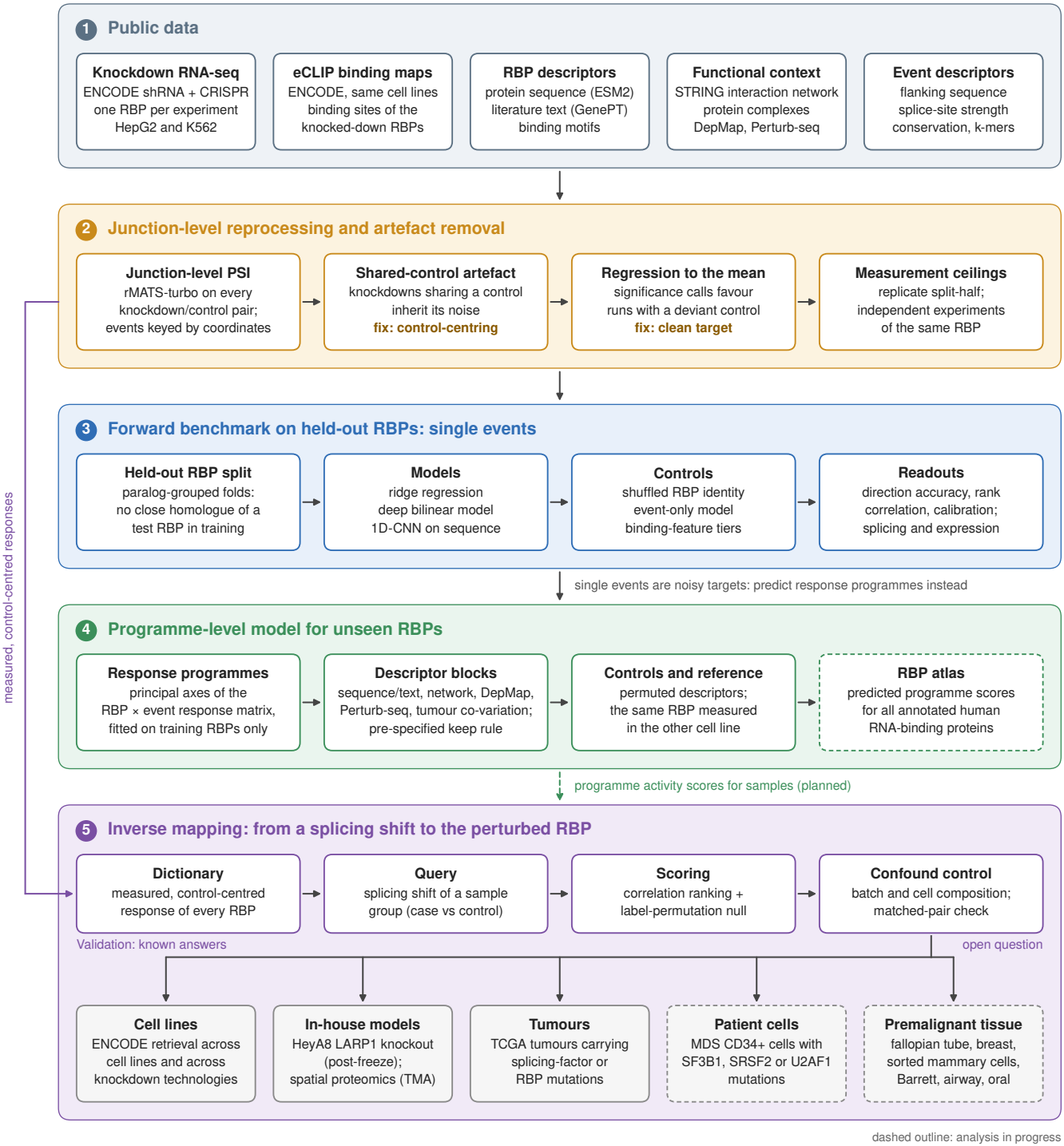
1 Public data
Knockdown RNA-seq
ENCODE shRNA + CRISPR
one RBP per experiment
HepG2 and K562
eCLIP binding maps
ENCODE, same cell lines
binding sites of the
knocked-down RBPs
RBP descriptors
protein sequence (ESM2)
literature text (GenePT)
binding motifs
Functional context
STRING interaction network
protein complexes
DepMap, Perturb-seq
Event descriptors
flanking sequence
splice-site strength
conservation, k-mers
2 Junction-level reprocessing and artefact removal
Junction-level PSI
rMATS-turbo on every
knockdown/control pair;
events keyed by coordinates
Shared-control artefact
knockdowns sharing a control
inherit its noise
fix: control-centring
Regression to the mean
significance calls favour
runs with a deviant control
fix: clean target
Measurement ceilings
replicate split-half;
independent experiments
of the same RBP
3 Forward benchmark on held-out RBPs: single events
Held-out RBP split
paralog-grouped folds:
no close homologue of a
test RBP in training
Models
ridge regression
deep bilinear model
1D-CNN on sequence
Controls
shuffled RBP identity
event-only model
binding-feature tiers
Readouts
direction accuracy, rank
correlation, calibration;
splicing and expression
single events are noisy targets: predict response programmes instead
4 Programme-level model for unseen RBPs
Response programmes
principal axes of the
RBP × event response matrix,
fitted on training RBPs only
Descriptor blocks
sequence/text, network, DepMap,
Perturb-seq, tumour co-variation;
pre-specified keep rule
Controls and reference
permuted descriptors;
the same RBP measured
in the other cell line
RBP atlas
predicted programme scores
for all annotated human
RNA-binding proteins
programme activity scores for samples (planned)
5 Inverse mapping: from a splicing shift to the perturbed RBP
Dictionary
measured, control-centred
response of every RBP
Query
splicing shift of a sample
group (case vs control)
Scoring
correlation ranking +
label-permutation null
Confound control
batch and cell composition;
matched-pair check
Validation: known answers
open question
Cell lines
ENCODE retrieval across
cell lines and across
knockdown technologies
In-house models
HeyA8 LARP1 knockout
(post-freeze);
spatial proteomics (TMA)
Tumours
TCGA tumours carrying
splicing-factor or
RBP mutations
Patient cells
MDS CD34+ cells with
SF3B1, SRSF2 or U2AF1
mutations
Premalignant tissue
fallopian tube, breast,
sorted mammary cells,
Barrett, airway, oral
measured, control-centred responses
dashed outline: analysis in progress